Kaustubh Hemant Sonawane

Bristol, UK | ksstubh.sonawane@gmail.com | +44 7388229743 | +91 8591441019 | www.github.com/Arishtanemi4 |
www.linkedin.com/in/kaustubhson4 | https://kaustubh-sonawane-portfolio.vercel.app

UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

Data Scientist with 3 years of industry experience in financial services (HDFC Asset Management), turning ambiguous business questions into analysis and decisions for senior, non-technical stakeholders, and an MSc in Data Science from the University of Bristol (awaiting result). Analysed customer buying patterns with **Market Basket Analysis**, built **Power BI** and **ReactJS** dashboards that moved financial analysts off Excel and saved them **10 hours a week**, and automated a rating-agency email pipeline handling over **300** emails a day that saves **60 analyst hours a month**. Co-built a **calibrated scoring and ranking algorithm** for an MSc dissertation with AstraZeneca, refined with scientist feedback, and placed 3rd in a Lloyds Bank-sponsored **credit card fraud detection** datathon. Worked directly with Compliance, Investment, Client Service and Marketing teams, shaping requirements and explaining results in plain language; twice recognised internally with Best Team Project (2024, 2025). Keen to bring analytical rigour and ownership to the Wallet, Payments & Commerce Analytics team on Apple Pay.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Analytics, Statistical Modelling & Insight

- Analyzed customer data using **Python Pandas** and identified customer buying patterns with **Market Basket Analysis** (**Apriori Algorithm**), handing the insights to the marketing team, who decided how to act on them; used **Pandas** in virtually every project, from data jobs to backend APIs.
- Built a **Python** pipeline on **AWS SageMaker** that turns over **300** daily emails from credit rating agencies into structured data, **fuzzy-matches** company names to the tracked list and sends analysts a consolidated daily summary, **saving 60 manual analyst hours a month**; upgraded parsing from **AWS Textract** to **Azure OpenAI** when a rating agency began sending emails as images, gaining much better accuracy.
- Designed and built **GenAI** applications end to end (**ReactJS**, **Python FastAPI**, **RAG** with **LlamaIndex** and **Azure OpenAI** GPT-3.5 & **Llama-3**) to answer department-specific queries over internal documents, with a separate **pgvector** database per department; first as the only engineer alongside my manager, later leading the team.
- Handled sensitive data responsibly: department-based document access, **role-based access control** with a **Python** authentication function against the company **LDAP** server, and Compliance review of every release; chose models on cost and fit, with **self-hosted Llama-3** for the largest department and stronger **Azure OpenAI** outputs for decision makers.
- Worked directly with business stakeholders in Client Service, Compliance, Investment and Marketing in a regulated financial-services setting to turn ambiguous requirements into working tools, released them to real users and improved them on their feedback, and explained results to non-technical audiences; acted as **de facto team lead** for up to 4 interns in my final year.

Reporting, Dashboards & Data Pipelines

- Developed **Power BI** dashboards for fortnightly credit report analysis for the investment team, migrating financial analysts from **Excel to Power BI** and improving analyst efficiency by **10 hours a week**.
- Recreated the dashboards in **ReactJS** inside the GenAI application, giving the wider organization access to the same data at lower cost than scaling **Power BI** licences.
- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** scripts ingesting the **Morningstar API** for every fund in the Indian mutual fund market (2023–2025), refreshed fortnightly and monthly and integrated with the company's legacy data; developed **Optimized SQL** queries, **SQL Functions** and **Procedures**.

Cloud & Deployment

- Designed the infrastructure architecture for internal analytics platforms and GenAI applications on **AWS** using core services such as **EC2**, **S3**, **ECR** and **Lambda**, with **Docker** microservices.
- Built one-click deployment to dev and prod using **Jenkins** and **GitLab** in a **CI/CD** pipeline, with separate Git repositories and dev/prod branches for frontend and backend.

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built. My focus: data preprocessing across heterogeneous scientific datasets (DepMap, GEO, HPA), the scoring algorithm and the RAG-based AI assistant for a tool that ranks cancer cell lines by target inclusion and exclusion genes; per-gene scoring across six evidence layers, weighted by correlation and calibrated across lineages, with the evidence behind each pick; validated and refined with feedback from AstraZeneca scientists.) | GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. Multi-phase architecture study comparing CNN/ViT/ResNet encoders and LSTM/attention decoders for generating natural-language image descriptions, evaluated with SBERT cosine, BLEU and ROUGE-L across five seeds and benchmarked against BLIP/SigLIP and commercial VLMs.) | GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **Swindon Borough Council Fuel-Poverty Hack:** household and neighbourhood-level fuel poverty data analysis tool built for the council's hackathon. | GitHub: https://github.com/Arishtanemi4/fuel-poverty.git

ACHIEVEMENTS

- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git

SKILLS

- **Tools:** Power BI, Jupyter, Anaconda, AWS (EC2, S3, ECR, Lambda, SageMaker, Textract), Git, GitLab, Jenkins, Docker, Azure OpenAI, LlamaIndex, pgvector, Claude Code, Nginx, Linux, VS Code.
- **Languages:** SQL, Python (pandas, scikit-learn, FastAPI), R (dplyr, tidyverse, ggplot2), NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector), SQLite